

Nathan Walsh

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Education

Purdue University

West Lafayette, IN

Bachelor of Science

May 2027

Majors: Data Science, Artificial Intelligence and Applied Statistics | **Minor:** Real Estate

GPA: 3.77/4.00

- **Purdue's Presidential Scholarship:** Merit-based award for high-achieving students
- **Dean's List and Semester Honors:** Selected every semester for outstanding academic achievement and cumulative GPA

Experience

College of Science Data Assistant

May 2025 - Present

Purdue University College of Science Office of Data Management

West Lafayette, IN

- Deliver reports and dashboards that translate academic and operational data into actionable insights, directly informing administrative decisions
- Collect, clean, and validate sensitive student and program data, ensuring accuracy and compliance with university standards
- Increase efficiency in workflows by documenting processes and leveraging tools such as Excel, Tableau, R, and Python

Undergraduate Data Science Research

August 2024 – December 2024

Research Team Member, The Data Mine Corporate Partners

West Lafayette, IN

- Built a machine learning model for BASF that leveraged historical forecast data to improve long-term weather pattern predictions, supporting data-driven strategies for optimizing grape yield
- Implemented a LSTM network with Tensorflow and Keras to capture temporal dependencies in weather and yield data
- Enhanced model performance through feature engineering and parameter tuning, which resulted in higher forecast accuracy and around 83% accuracy when predicting grape yield

Projects

MLB Single Game Outcome Predictor

January 2025 - March 2025

Lead Developer, Group Project with Sports Analytics Club, Purdue University

West Lafayette, IN

- Analyzed large-scale historical baseball performance datasets to engineer predictive features and key indicators of the winner of a single game
- Trained a logistic regression model in Python with Scikit-learn to classify home wins vs losses which achieved an 8% increase in prediction accuracy over the baseline
- Translated model results into interpretable insights for a presentation competition in Sports Analytics Club

Activities and Involvement

Purdue Real Estate Society

January 2026 - Present

Member, Professional Development and Academic Society

West Lafayette, IN

- Engage in workshops and industry panels focused on commercial real estate fundamentals and career paths
- Network with real estate professionals, alumni, and peers to gain exposure and exchange perspectives regarding emerging industry standards

Purdue Hurling Club

August 2024 - Present

Member, Traditional Irish Sports club

West Lafayette, IN

- Participate in hurling, a fast-paced traditional Irish field sport, enhancing physical fitness and teamwork skills
- Collaborate with teammates to strategize and compete in club matches, fostering communication and leadership on and off the field
- National Collegiate Hurling Champions 2025 and 2026, Dr. David Pinkston Player of the Year 2025

College of Science Ambassador

October 2024 - Present

Purdue University

West Lafayette, IN

- Serve as a representative of the College of Science, engaging with prospective students and families during campus visits and outreach events
- Communicate academic, research, and student life opportunities with clarity and enthusiasm, resulting in many students from different backgrounds having positive first impressions of Purdue

Technical Skills

Finance and Analytics: Microsoft Excel, Tableau

Data Science Frameworks: Pandas, NumPy, Matplotlib, Scikit-learn, TensorFlow, Keras

Programming Languages: Python, R, Java, SQL, SAS